AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2014

Chemistry Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

There are three students A, B and C with a shared interest in Chemistry. However, they like to specialize in different branches of Chemistry studying at the University level. Identify in which branch of Chemistry each student will enroll as per his/her interest.

S. NO.	Interest	Branch of Chemistry
1.	**Student A** is interested in studying about the laws and principles governing the combination of atoms and molecules.	
2.	**Student B** likes to study about the compounds and metabolism of a living body.	
3.	**Student C** wants to learn about the changes occurring in the nuclei of atoms, accompanied by the emission of invisible radiations.	

Q.2. (Total 4 Marks)

a. Rutherford carried out a gold metal foil experiment based on which he made a conclusion that an atom consists of two parts: 1) Nucleus 2) Extra nuclear part. However, there were certain weaknesses in his atomic model which led to further experimentations by other scientists.

Mention TWO defects in Rutherford's atomic model. (2 Marks)

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b. Draw an electronic dot and cross structure of calcium chloride (ionic compound) showing the arrangement of valence electrons with correct charges on the ions. [Hint: Atomic number of Ca = 20 and Cl = 17] (2 Marks)

Space for Structure:

Q.3. (Total 5 Marks)

An electronic configuration of an element is given below.

Electronic configuration: $1s^2, 2s^2, 2p^6, 3s^2, 3p^3$

a. Predict the group and period of the element. (2 Marks)

Group: _____

Period: _____

b. On what basis are elements arranged in the modern periodic table? Give any TWO differences between a group and a period. (3 Marks)

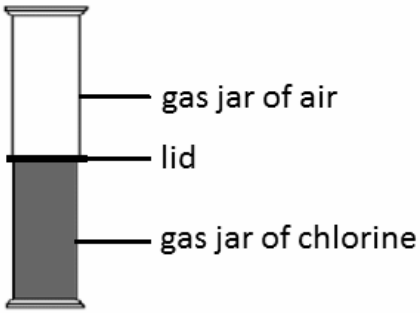
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_____

S. No.	Group	Period
1.		
2.		

Q.4. (Total 4 Marks)

a. A gas jar of air is placed over another gas jar containing a dense gas, chlorine. Both jars are separated by a lid as shown in the figure.



i. Name the phenomenon that will occur after the lid is removed. (1 Mark)

_____

_____

ii. Give a reason using the idea of particle why the phenomenon that you have identified in part (i) will occur. (1 Mark)

_____

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b. A deep blue copper sulphate solution turns pale blue when water is added.

Name and describe the phenomenon that causes the change in colour from deep blue to pale blue. (2 Marks)

_____

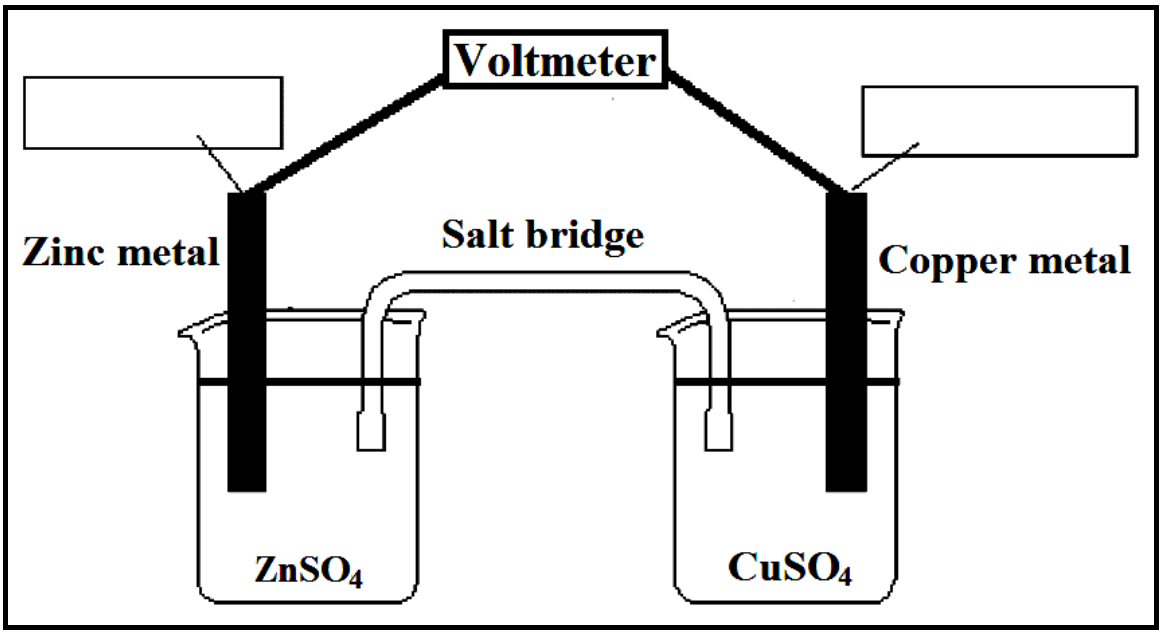
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Q.5. (Total 5 Marks)

a. A sketch of Daniel cell is given below.



i. Label the cathode and anode in the given diagram. (1 Mark)

ii. Show the flow of current with the help of arrows in the given diagram. (1 Mark)

iii. Identify at which electrode the process of oxidation occurs. (1 Mark)

b. NaOH is observed to be a strong electrolyte that conducts electricity.

i. What makes sodium hydroxide conduct electricity in aqueous solutions? (1 Mark)

_____

_____

_____

ii. Write a chemical equation to support your answer to part (i). (1 Mark)

_____

_____

Q.6. (Total 4 Marks)

Magnesium and calcium show similar chemical behaviour due to which they are placed in the same group of the periodic table. Their electronic configurations are as follows:

$_{12}Mg$	$1s^2,2s^2,2p^6,3s^2$
$_{20}Ca$	$1s^2,2s^2,2p^6,3s^2,3p^6,4s^2$

a. State the group to which Mg and Ca belong. (1 Mark)

b. Describe why the ionization energy of calcium is less than that of magnesium. (2 Marks)

c. What will be the product of the given reaction between magnesium oxide and water?

$MgO + H_2O \rightarrow ?$ (1 Mark)

Q.7. (Total 8 Marks)

EITHER

a. Identify and describe the types of chemical reaction shown in each of the given equations.

i. $2Mg + O_2 \rightarrow 2MgO$

ii. $H_2S + Cl_2 \rightarrow 2HCl + S$

iii. $2Zn(NO_3)_2 \xrightarrow{\Delta} 2ZnO + 4NO_2 + O_2$

iv. $C_2H_5OH + 3O_2 \xrightarrow{\Delta} 2CO_2 + 3H_2O$

OR

b. Give any ONE example of radioactive isotopes for each of the following fields and also describe the use of it.

i. Radiotherapy

ii. Medicine

iii. Archaeology and Geology

iv. Power generation in nuclear reactors

Q.8. (Total 7 Marks)

EITHER

a. Write any FIVE differences between amorphous and crystalline solids giving ONE example of each.

S. No.	Amorphous Solid	Crystalline Solid
1.		
2.		
3.		
4.		
5.		
Example:		

OR

b. Describe the manufacturing of sodium metal from fused sodium chloride using Down's cell. Write in detail the construction and working of Down's cell giving necessary chemical equations.

END OF PAPER

Please use this page for rough work